HUNT-SAGE · THREAT HUNTING CHEAT SHEET

Windows Security Event IDs for Hunters

The events that matter, what they mean, and which technique they expose.

github.com/iamjammie/hunt-sage · iamjammie.github.io/hunt-sage · Built 2026-10-31

Events marked **(audit)** require the matching Advanced Audit Policy subcategory. See Data & telemetry.

Authentication and logon

ID	Meaning	Hunting use
4624	Successful logon	Logon type, source IP, auth package, `LogonId` to link activity; lateral movement
4625	Failed logon	Brute force, password spray; `Status` / `SubStatus` codes below
4634 / 4647	Logoff / user-initiated logoff	Session duration
4648	Logon with explicit credentials	RunAs, PsExec `-u`, pass-the-hash tooling, lateral movement
4672	Special privileges assigned	Admin logons; new accounts with privileges
4776	NTLM credential validation (DC)	NTLM usage, spray against DCs
4768	Kerberos TGT requested	AS-REP roasting (pre-auth type 0), RC4 downgrade, unusual sources
4769	Kerberos service ticket requested	Kerberoasting (RC4 `0x17`), ticket requests by service
4770 / 4771	Ticket renewed / pre-auth failed	4771 spikes = spray via Kerberos
4778 / 4779	RDP session reconnected / disconnected	RDP session tracing; `ClientName` and source
4800 / 4801	Workstation locked / unlocked	User presence

Logon types (4624/4625)

Type	Name	Notes
2	Interactive	Console
3	Network	SMB, WinRM, many remote tools
4	Batch	Scheduled tasks
5	Service	Service start
7	Unlock	
8	NetworkCleartext	IIS basic auth, etc.; credential exposure
9	NewCredentials	`runas /netonly`; Mimikatz `sekurlsa::pth`
10	RemoteInteractive	RDP
11	CachedInteractive	Offline domain logon

Common failure codes (4625 `Status` / `SubStatus`)

Code	Meaning
`0xC000006D`	Bad username or authentication info
`0xC000006A`	Right username, wrong password
`0xC0000064`	Username does not exist (spray or enumeration)
`0xC0000234`	Account locked out
`0xC0000072`	Account disabled
`0xC0000070`	Logon time restriction / workstation restriction
`0xC0000193` / `0xC0000071`	Account expired / password expired

Kerberos encryption types (4768/4769)

Value	Cipher	Note
`0x17`	RC4-HMAC	Preferred by roasting tools; rare in modern AES-only estates
`0x12`	AES256-CTS-HMAC-SHA1-96	Normal
`0x11`	AES128-CTS-HMAC-SHA1-96	Normal
`0x3` / `0x1`	DES-CBC-MD5 / DES-CBC-CRC	Legacy; investigate

Process, scheduled tasks and services

ID	Meaning	Hunting use
4688	Process created **(audit; enable command line)**	Execution chains when Sysmon is absent
4689	Process exited	Duration of short-lived tools
4698 / 4699 / 4700 / 4701 / 4702	Scheduled task created / deleted / enabled / disabled / updated **(audit)**	Persistence, remote execution
4697	Service installed (Security)	Persistence, PsExec-like
7045 (System)	New service installed	Same, higher coverage; image path
7034 / 7036 / 7040 (System)	Service crashed / state change / start type changed	Tamper, defence evasion

Accounts and groups

ID	Meaning
4720	User account created
4722 / 4725	Account enabled / disabled
4723 / 4724	Password change / reset attempt
4726	Account deleted
4728 / 4729	Member added / removed: global security group
4732 / 4733	Member added / removed: local security group (e.g. Administrators)
4756 / 4757	Member added / removed: universal group
4738	User account changed (e.g. `UserAccountControl` flags such as pre-auth off)
4740	Account locked out
4767	Account unlocked
4794	DSRM password set attempt

Directory service and Active Directory

ID	Meaning	Hunting use
4662	Operation performed on an object **(audit)**	DCSync via replication GUIDs
5136 / 5137 / 5141	Directory object modified / created / deleted **(audit)**	ACL abuse, GPO tampering, AdminSDHolder changes, shadow credentials (`msDS-KeyCredentialLink`)
4670	Permissions on an object changed	Delegation backdoors
4741 / 4742	Computer account created / changed	RBCD, rogue machine accounts (`ms-DS-MachineAccountQuota` abuse)
4928 / 4929	AD replica source naming context established/removed	Rogue DC (DCShadow)

Network shares and object access

ID	Meaning	Hunting use
5140	Network share accessed **(audit)**	`ADMIN$` , `C$` , `IPC$` access from unusual sources

ID	Meaning	Hunting use
5145	Network share object checked	File-level view; PsExec/payload copy to `ADMIN$`
5142-5144	Share created / modified / deleted	Staging shares
4663 / 4656	Object access attempt / handle requested	File/registry/LSASS SACL auditing
4657	Registry value modified	Persistence if SACL set

Policy, log and system integrity

ID	Meaning
1102	Security log cleared
104 (System)	Other event log cleared
4719	Audit policy changed (evasion)
4616	System time changed
4624 + 4672 sequences	Admin logon pairs for baselining
6416	New external device recognised (USB)
4964	Special groups assigned to new logon
1100	Event logging service shut down

Other logs worth ingesting

Log	Highlights
`Microsoft-Windows-PowerShell/Operational`	4104 script block, 4103 module logging
`Microsoft-Windows-WinRM/Operational`	91, 168: WinRM session activity
`Microsoft-Windows-TerminalServices-RemoteConnectionManager/Operational`	1149: RDP auth succeeded (pre-logon), source IP
`Microsoft-Windows-TerminalServices-LocalSessionManager/Operational`	21, 22, 23, 24, 25: session logon/logoff/reconnect
`Microsoft-Windows-TaskScheduler/Operational`	106, 140, 141, 200, 201: task registered/updated/removed/run
`Microsoft-Windows-WMI-Activity/Operational`	5857, 5860, 5861: WMI provider load and event subscription activity
`Microsoft-Windows-Windows Defender/Operational`	1116/1117 detections; 5001, 5007, 5010, 5012: tampering
`Microsoft-Windows-Bits-Client/Operational`	3, 59, 60, 61: BITS jobs
`Microsoft-Windows-DNS-Client/Operational`	3008, 3020: client DNS queries (without Sysmon)
`Microsoft-Windows-SMBServer/Security`	1006, 1007...: SMB server security events
`Microsoft-Windows-Sysmon/Operational`	See Sysmon sheet

Quick hunts by event

```
Password spray       4625 (0xC000006A / 0xC0000064) from one source to many users; 4771 on DCs
Pass-the-hash        4624 type 3 NTLM, KeyLength 0; type 9 + seclogo
Kerberoasting        4769, 0x17, user-owned SPNs, many distinct SPNs per requester
DCSync               4662 with 1131f6aa-/1131f6ad-... GUIDs from non-DC
Remote exec (PsExec) 5145 ADMIN$ write -> 7045/4697 service -> child of services.exe
Scheduled task       4698 (Task XML shows command); 4699 deletion after use = cleanup
Log clearing         1102 / 104; 1100; 4719
Privilege change     4728/4732/4756 for admin groups; 4672 for new accounts
```